

## Chapter 8 Power BI Dashboard Design

To communicate the results of the Airbnb market analysis, four interactive Power BI dashboards were developed. Each dashboard focuses on a specific business objective, transforming analytical results into practical insights for market evaluation, pricing strategy, demand understanding, and investment decision-making. The dashboards combine KPI cards, maps, comparison charts, scatter plots, and ranking tables to allow users to explore differences between cities and identify potential opportunities.

### Dashboard 1 — Airbnb Global Market Overview

#### Objective

The dashboard summarises the Airbnb marketplace across 280K listings and 182K hosts. This dashboard answers: "What does the Airbnb market look like across different locations?"



#### Visuals and Analysis

##### ✓ KPI Cards

The dashboard begins with KPI cards summarising key market indicators:

| KPI            | Value | Interpretation                                                             |
|----------------|-------|----------------------------------------------------------------------------|
| Total Listings | 280K  | Indicates a large and diverse accommodation market across multiple cities. |
| Total Hosts    | 182K  | Shows strong participation from both new and experienced hosts.            |

|                            |          |                                                                                  |
|----------------------------|----------|----------------------------------------------------------------------------------|
| <b>Average Price</b>       | \$115.03 | Represents the overall average nightly price across markets.                     |
| <b>Average Rating</b>      | 94.25    | Indicates generally high customer satisfaction.                                  |
| <b>Average Value Score</b> | 0.85     | Suggests listings provide relatively strong value considering price and quality. |

The high average rating combined with a strong value score indicates that Airbnb maintains a competitive customer experience across different markets. However, pricing may vary significantly between cities, suggesting opportunities for location-based investment strategies.

### ✓ *Airbnb Price and City Distribution Map*

The geographic distribution map illustrates variations in Airbnb pricing performance across international cities. The analysis shows that Airbnb markets differ considerably depending on location, tourism demand, and local economic conditions.

Premium pricing markets are mainly concentrated in major tourism and business destinations. Cities such as **Sydney, New York, Paris, and Cape Town** demonstrate higher average accommodation prices, suggesting stronger pricing power and potentially higher revenue opportunities for property owners.

In contrast, cities such as **Mexico City and Bangkok** have lower average prices, indicating more affordable accommodation markets. While these markets generate lower revenue per booking, they may provide investment opportunities due to potentially lower property acquisition costs and opportunities for future market growth.

Overall, the geographic analysis highlights that high accommodation prices alone do not determine investment attractiveness. Investors should also consider market competition, operating costs, customer demand, and occupancy potential.

### ✓ *City-Level Pricing Analysis*

The average price comparison identifies significant differences in Airbnb pricing strategies between cities.

Sydney records the highest average Airbnb price at approximately \$153 per night, followed by Cape Town (\$149) and New York (\$143). These cities represent premium accommodation markets where hosts can achieve higher revenue per booking. However, high-price markets may also involve greater competition and higher property costs.

Other major tourism destinations, including Rio de Janeiro (\$141) and Paris (\$129), also demonstrate strong pricing performance. These results suggest that cities with strong tourism activity and international visitor demand tend to support higher Airbnb prices.

On the other hand, Mexico City has the lowest average price among analysed markets at approximately \$66 per night, followed by Hong Kong at \$97 per night. Although lower prices may reduce revenue per booking, these markets could still be attractive if supported by high occupancy rates and lower operating expenses.

The findings suggest that investors should evaluate both revenue potential and market accessibility when selecting Airbnb investment locations.

### ✓ *Preferred Room Type Analysis*

The room type analysis demonstrates clear customer preferences within the Airbnb marketplace. Entire-place accommodations represent the largest segment, accounting for approximately 182,000 listings (65.07%). Private rooms represent the second-largest category with 87,000 listings (31.1%), while hotel rooms and shared rooms contribute only a small proportion of total supply.

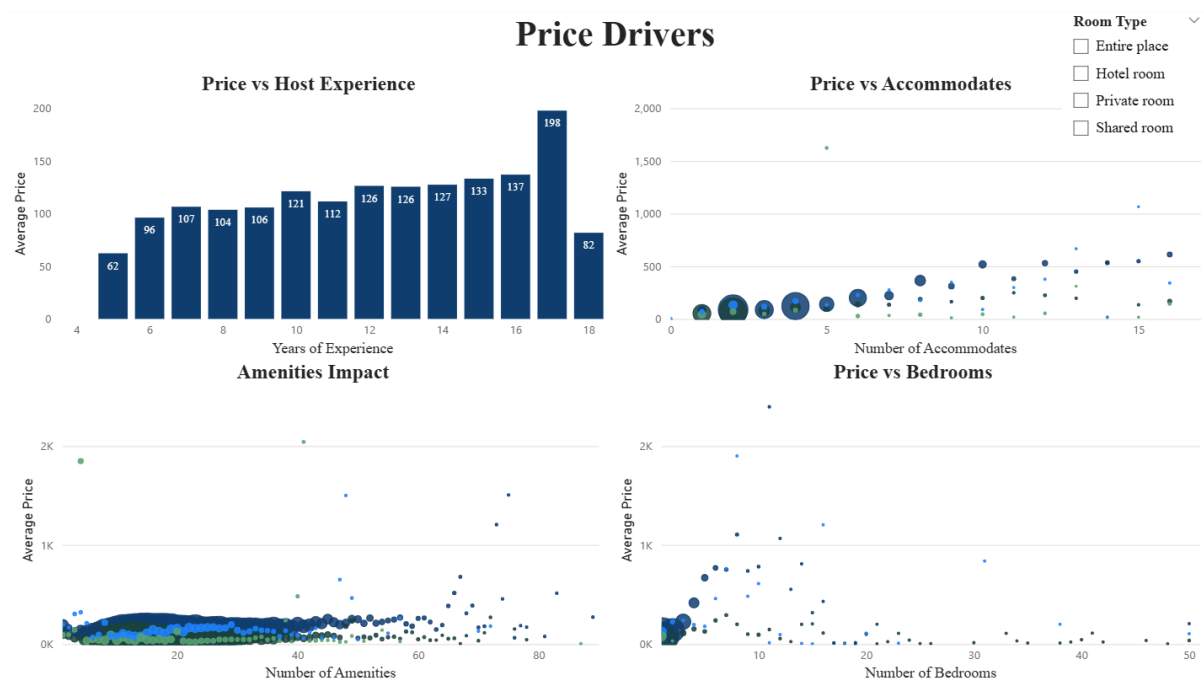
The dominance of entire-place listings indicates that Airbnb customers increasingly prefer private and independent accommodation experiences rather than shared facilities. This trend may be driven by factors such as privacy, flexibility, and convenience, particularly among families, business travellers, and longer-term visitors.

For investors, this suggests that entire-property investments may have stronger market demand. However, these properties typically require higher initial capital investment and therefore should be evaluated alongside expected revenue and operating costs.

## **Dashboard 2 — Price Drivers**

### **Objective**

The Price Drivers dashboard investigates the factors influencing Airbnb pricing behaviour. The key business question addressed is: "Why are some Airbnb listings more expensive than others?"



## Visuals and Analysis

### ✓ *Price vs Host Experience*

The analysis shows a **positive relationship between host experience and average listing price**. In other words, listings managed by hosts with more years of experience generally achieve higher average prices. In particular, hosts with **5–10 years of experience** have average prices ranging approximately from **\$60 to \$120** while listings from hosts with **12–17 years of experience** show higher average prices, reaching around **\$125–\$198**. Besides, the highest average price is observed among hosts with approximately **17 years of experience**, with an average price close to **\$198**. Experienced hosts may achieve higher prices due to several factors, such as a better understanding of market demand and pricing strategies, a stronger reputation and accumulated guest reviews, higher-quality property management practices, and greater ability to optimise listing presentation, services, and guest experience. However, host experience alone does not determine price, as other property-level factors such as location, property type, and size may also contribute significantly. For investors, properties managed by experienced hosts may indicate higher operational maturity and stronger revenue potential while new hosts should focus on improving service quality and building review history to increase future pricing power.

### ✓ *Price vs Accommodates*

Property capacity is one of the strongest price drivers because larger properties typically provide more bedrooms and living space, the ability to host groups and families, and higher overall customer value. The presence of extreme high-price listings suggests that large-capacity properties may represent premium segments, especially when combined with desirable locations or additional facilities. The

result shows that the relationship between the **number of accommodations and listing price** demonstrates a clear **positive** trend, with smaller properties accommodating 1–4 guests generally having lower average prices, properties accommodating 6–10 guests showing a significant increase in average price, and larger properties accommodating 10+ guests achieving some of the highest prices, with several listings exceeding \$500–\$1,000. This demonstrates that investors seeking higher revenue opportunities should consider properties with greater accommodation capacity. However, larger properties also require higher acquisition and maintenance costs, so profitability should be evaluated using revenue potential rather than price alone.

#### ✓ *Price vs Amenities*

The relationship between the number of amenities and average price appears relatively **weak** for most properties. Most listings provide approximately 10–30 amenities, with average prices concentrated below \$300. Some outliers with 50+ amenities reach prices above \$1,000, although these cases are limited. This result shows that the greater number of amenities does not automatically result in higher prices. Instead, the type and quality of amenities may be more important than the total count.

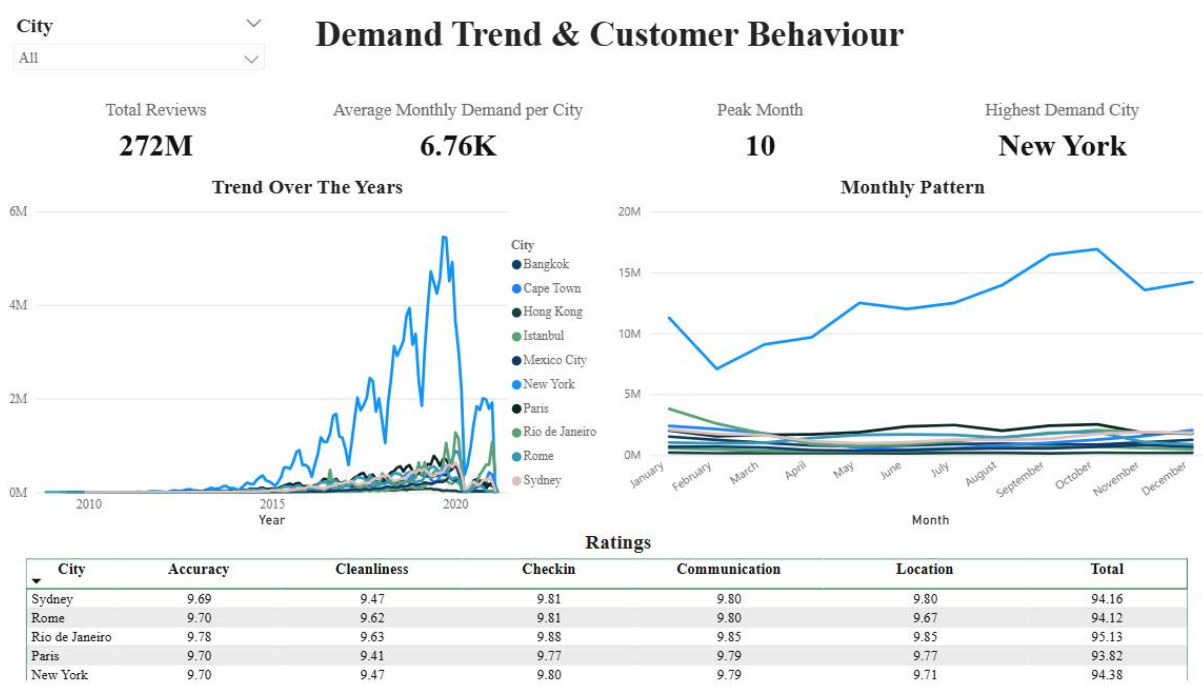
#### ✓ *Price vs Bedrooms*

Bedroom count is an important indicator of property size and market positioning. Larger properties can attract group travellers, corporate bookings, family holidays, or premium customers. The number of bedrooms shows a **positive but highly variable** relationship with price, in which listings with 1–3 bedrooms represent the majority of the market and generally have lower average prices while properties with 5+ bedrooms tend to achieve higher prices. Some large properties with 10–30 bedrooms demonstrate significantly higher prices, exceeding \$1,000–\$2,000. However, the wide variation suggests that bedroom count alone is insufficient to explain pricing differences. Location, property type, and additional services likely influence the final price.

### **Dashboard 3 — Demand Analysis and Customer Behaviour**

#### **Objective**

The Demand Analysis dashboard evaluates traveller activity and market trends over time. The dashboard answers: "When is Airbnb demand highest, and which cities attract the most travellers?"



## Visuals and Analysis

### ✓ KPI Cards

| KPI                             | Value        | Interpretation                                                                                                                                  |
|---------------------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Total Reviews                   | 272M         | Reflects a very large, mature review base built up across all ten cities over the full period covered.                                          |
| Average Monthly Demand per City | 6.76K        | Represents typical monthly review activity per city; the modest average shows demand is concentrated in a few cities rather than spread evenly. |
| Peak Month                      | 10 (October) | Indicates demand consistently peaks in Q4, likely driven by autumn leisure and pre-holiday travel.                                              |
| Highest Demand City             | New York     | Confirms New York as the single largest market by review volume, well ahead of all other cities tracked.                                        |

The gap between the very high Total Reviews figure and the modest Average Monthly Demand per City shows that overall volume is driven by a small number of high-performing cities rather than uniform growth everywhere. The October peak points to a recurring Q4 demand spike, and New York's clear lead as Highest Demand City marks it as the dominant market in the dataset.

### ✓ Yearly Demand Trend

The yearly trend chart tracks review growth for each city from 2010 to 2021. Demand across all cities was minimal before 2014, then grew steadily through 2015–2018 before accelerating sharply. New

York separates from the rest of the group from around 2016 onward, reaching peaks two to three times higher than any other city by 2019.

All cities, including New York, show a sharp decline toward the end of the period. This drop is consistent across every market at the same time, suggesting a shared external cause rather than a decline specific to any one city. The decrease is primarily attributed to the impact of the COVID-19 pandemic, which significantly disrupted global travel, reduced tourism activity, and introduced restrictions on domestic and international movement. As a result, accommodation demand declined sharply in 2020, leading to fewer bookings and, consequently, fewer customer reviews recorded across all Airbnb markets. This trend highlights the strong dependency of short-term rental demand on external factors such as travel conditions and economic disruptions.

#### ✓ *Monthly Seasonal Pattern*

The observed seasonal variation is likely influenced by differences in tourism cycles, climate conditions, public holidays, and major local events. For example, cities with strong summer tourism seasons may experience higher demand during warmer months, while destinations hosting cultural festivals, business events, or international conferences may see temporary increases during specific periods. These local factors contribute to differences in the timing and intensity of demand peaks across markets.

The chart uses a 20 million review scale, which clearly demonstrates the large gap between New York and the other markets. Due to its significantly higher review volume, New York appears as a standalone market, while the remaining cities are grouped closely together below approximately 2 million reviews. This highlights New York's dominant position and its substantial contribution to overall Airbnb activity within the dataset.

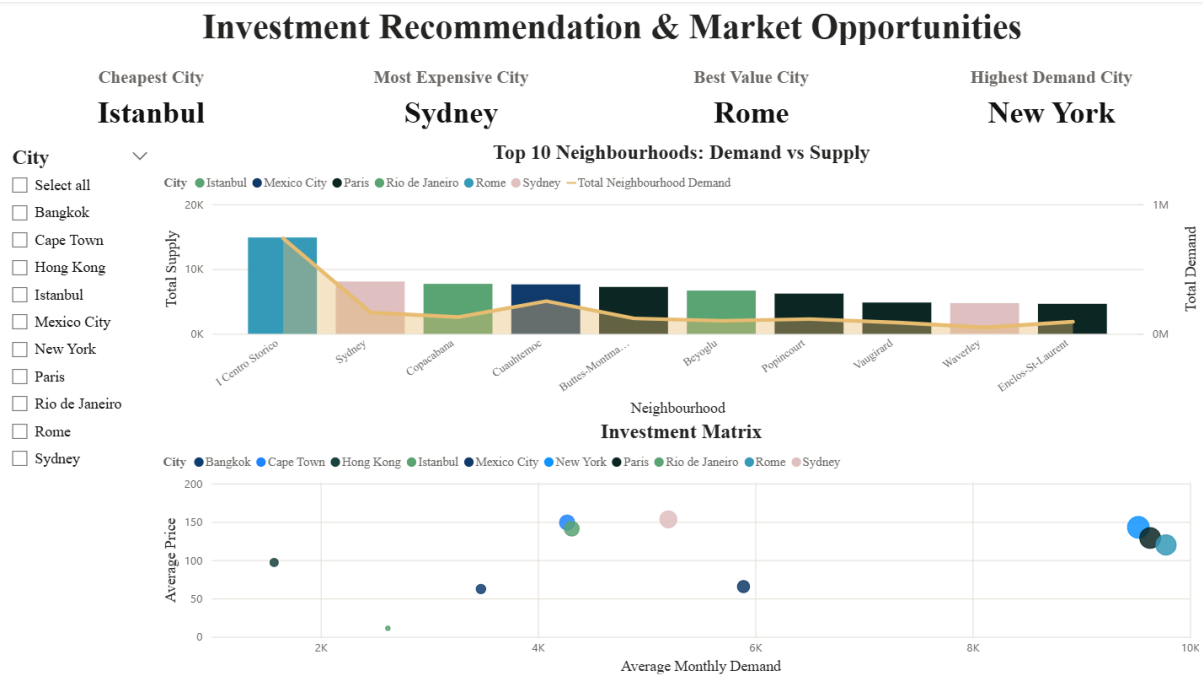
#### ✓ *Rating Performance Table*

The ratings table breaks guest satisfaction down into five categories, including accuracy, cleanliness, check-in, communication, and location. Rio de Janeiro has the highest overall rating (95.13), while New York, despite leading in demand, ranks only mid-table (94.38) with the lowest cleanliness score among the top cities shown. Cleanliness is consistently the weakest category across all five cities, suggesting it is a shared area for improvement rather than an issue limited to one market.

### **Dashboard 4 — Investment Recommendation & Market Opportunities**

#### **Objective**

The Investment Analysis dashboard combines supply, demand, pricing, and quality indicators to support investment decision-making. The key question addressed is: "Which cities and neighbourhoods provide the most attractive investment opportunities?"



Visuals and Analysis

✓ *KPI Cards*

| KPI                 | Value    | Interpretation                                                                                                                                 |
|---------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Cheapest City       | Istanbul | The lowest average nightly price of any city tracked, pointing to a lower cost of entry for hosts and more budget-friendly options for guests. |
| Most Expensive City | Sydney   | The highest average nightly price, indicating strong pricing power but also a higher-cost, more premium market.                                |
| Best Value City     | Rome     | Offers the strongest balance of price against quality and demand, making it the most attractive city on a value-for-money basis.               |
| Highest Demand City | New York | New York remains the largest market by booking/review volume, consistent with the demand dashboard.                                            |

Istanbul and Sydney represent the two extremes of the pricing range, highlighting the substantial difference in average accommodation costs between markets. This variation can be influenced by several factors, including differences in local economic conditions, tourism demand, accommodation supply, and currency exchange rates when prices are converted into USD for comparison.

Rome being identified as the Best Value City, rather than simply the lowest-priced market, suggests that value is driven by a balance between affordability, customer experience, and demand levels. This indicates that a lower price alone does not necessarily represent the strongest investment opportunity;



instead, markets that achieve a favourable combination of competitive pricing, high-quality listings, and strong customer interest may provide greater value.

New York continues to demonstrate the highest demand among all analysed cities, reinforcing its position as the largest Airbnb market in the dataset. However, its leading demand level does not directly translate into the lowest prices or highest value score, demonstrating that market size, affordability, and overall value are separate factors that should be considered independently when evaluating investment opportunities.

### ✓ ***Top 10 Neighbourhood Supply and Demand Analysis***

This chart evaluates the relationship between accommodation supply (represented by the bars on the left axis) and customer demand (represented by the line on the right axis) across the ten neighbourhoods with the highest listing volumes.

I Centro Storico in Rome is a clear outlier, recording the highest number of listings as well as significantly stronger demand compared with all other neighbourhoods. The sharp increase in demand within this area suggests that it is a major contributor to Rome's overall market performance and helps explain why Rome achieved the highest Best Value City ranking in the analysis. The combination of strong visitor interest and substantial supply indicates that this neighbourhood benefits from both market attractiveness and established accommodation availability.

In comparison, the remaining neighbourhoods, including Sydney's Waverley, Rio de Janeiro's Copacabana, Mexico City's Cuauhtemoc, Paris's Buttes-Montmartre and Popincourt, Istanbul's Beyoglu, Vaugirard, and Enclos-St-Laurent, operate within a much narrower range. Their demand levels remain relatively similar and stable despite differences in listing supply. This suggests that demand is more evenly distributed across these locations rather than being heavily concentrated in a single neighbourhood. For investors, this highlights the importance of identifying specific high-performing areas, as city-level performance may be driven by only a small number of attractive neighbourhoods.

### ✓ ***Investment Matrix: Demand vs Price Opportunity***

The Investment Matrix compares average monthly demand against average listing price for each city, with bubble size representing the overall market scale. This visualisation helps identify markets that offer attractive combinations of customer interest and revenue potential.

Cape Town, Paris, and Rome appear in the upper-right section of the matrix, indicating strong demand alongside relatively high average prices. These markets represent potentially attractive investment opportunities because they combine customer willingness to pay with consistent market activity.

Rome's position within this high-demand and high-price group further supports its Best Value City ranking, as it achieves a strong balance between revenue potential and customer appeal.

New York, Istanbul, and Sydney demonstrate higher average prices but comparatively lower demand levels than the leading group. This suggests that while these markets may generate higher revenue per booking, they may also face stronger competition, higher operating costs, or a smaller demand base relative to their price levels.

Meanwhile, Hong Kong, Bangkok, and Mexico City occupy a middle position, offering moderate demand with lower average prices. These markets may provide opportunities for investors seeking more affordable entry points, although their revenue potential may be limited compared with premium markets. Rio de Janeiro appears in the least favourable position, with both lower demand and lower prices, indicating weaker investment attractiveness under the current metrics.

Overall, the Investment Matrix reinforces the earlier KPI findings by demonstrating that the strongest opportunities are not necessarily the largest or most expensive markets. Instead, cities that achieve a balance between demand strength, pricing power, and market scale provide the most attractive conditions for investment.